Evaluating multi-season occupancy models with autocorrelation fitted to heterogeneous datasets

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Supporting Information E - Evaluation of model misspecification, identifiability of $\psi_{it} \times p_{it}$, and analysis of simulated data using NNGP = 15 spatial neighbors.

Model missispecification

We evaluated misspecification by simulating occupancy and detection probabilities with the probit link function, and analyzed the simulated data with the logit link function. Missispecification was applied to the baseline scenario of Doser & Stoudt (2024) (our study 1 - scenario 0), to our scenario of total overlap of covariates (study 2 - scenario 2), and the scenarios with spatial and temporal cluster of data (study 3 - scenario 2 and study 3 - scenario 3). The results resemble those reported in the main text.

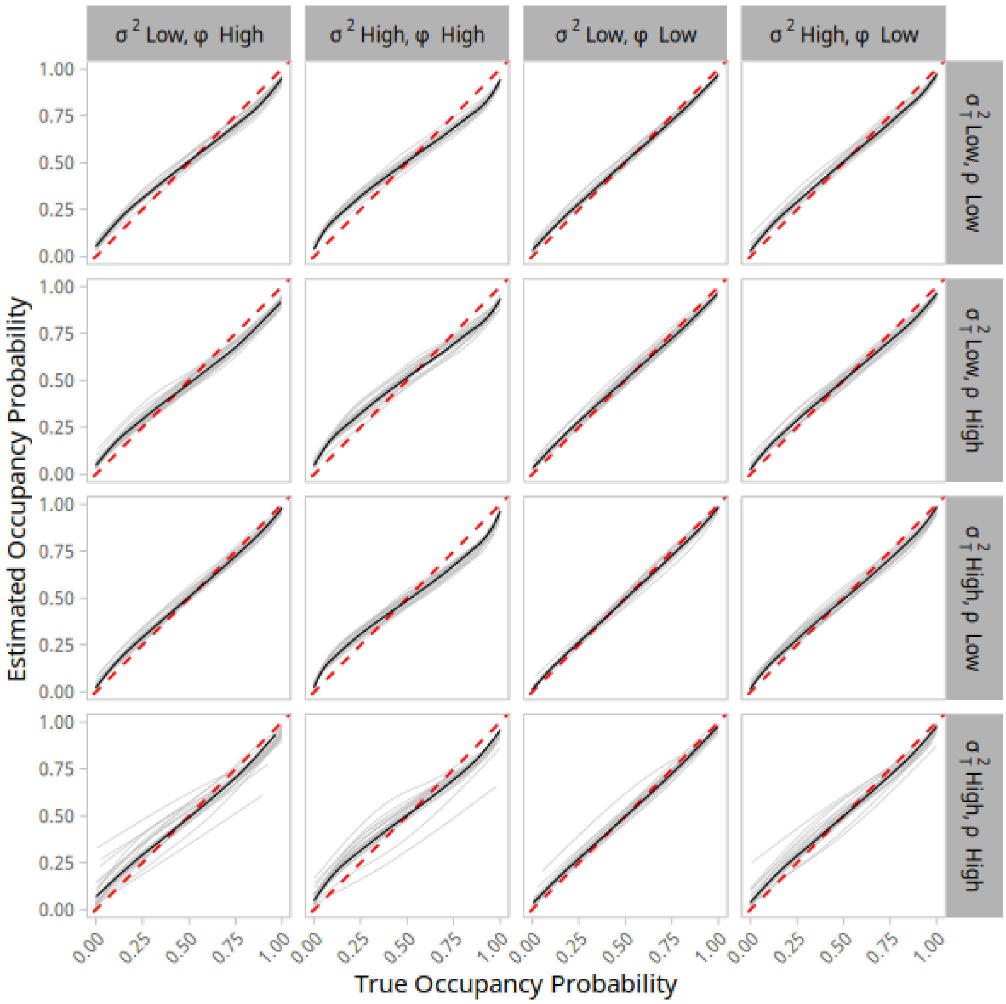

Fig. E.1: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) under a missispecified occupancy model. Results for the study 1-scenario 0 (baseline). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 20 simulated datasets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

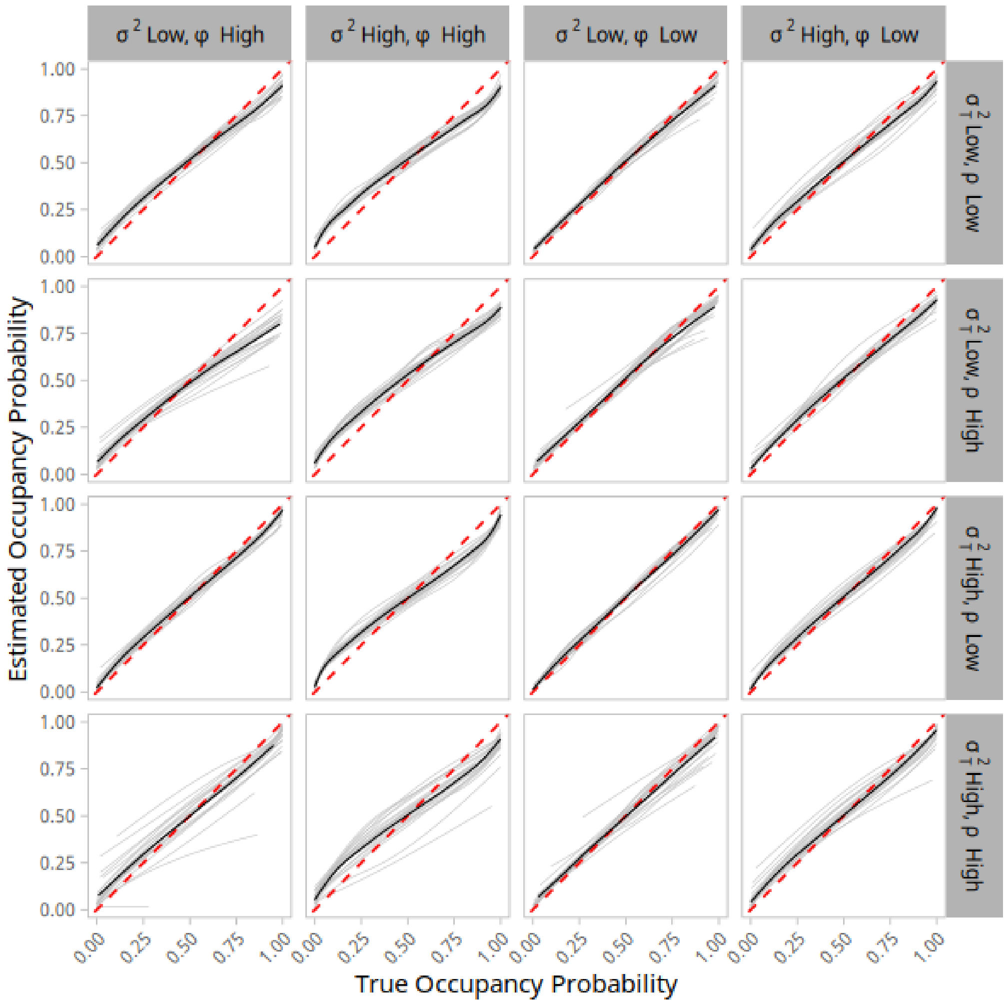

Fig. E.2: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) under a missispecified occupancy model. Results for the study 2-scenario 2 (overlap of covariates). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 20 simulated datasets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

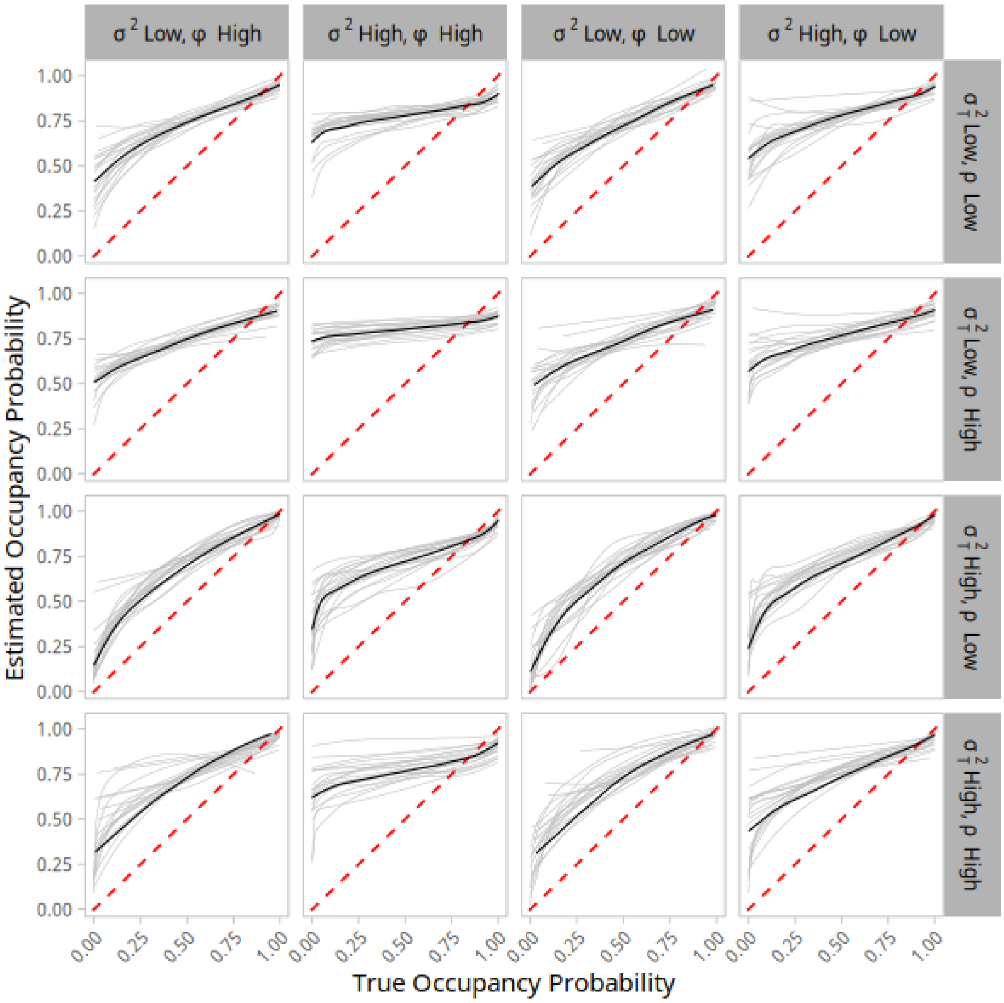

Fig. E.3: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) under a missispecified occupancy model. Results for the study 3-scenario 2 (observation spot). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 20 simulated datasets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

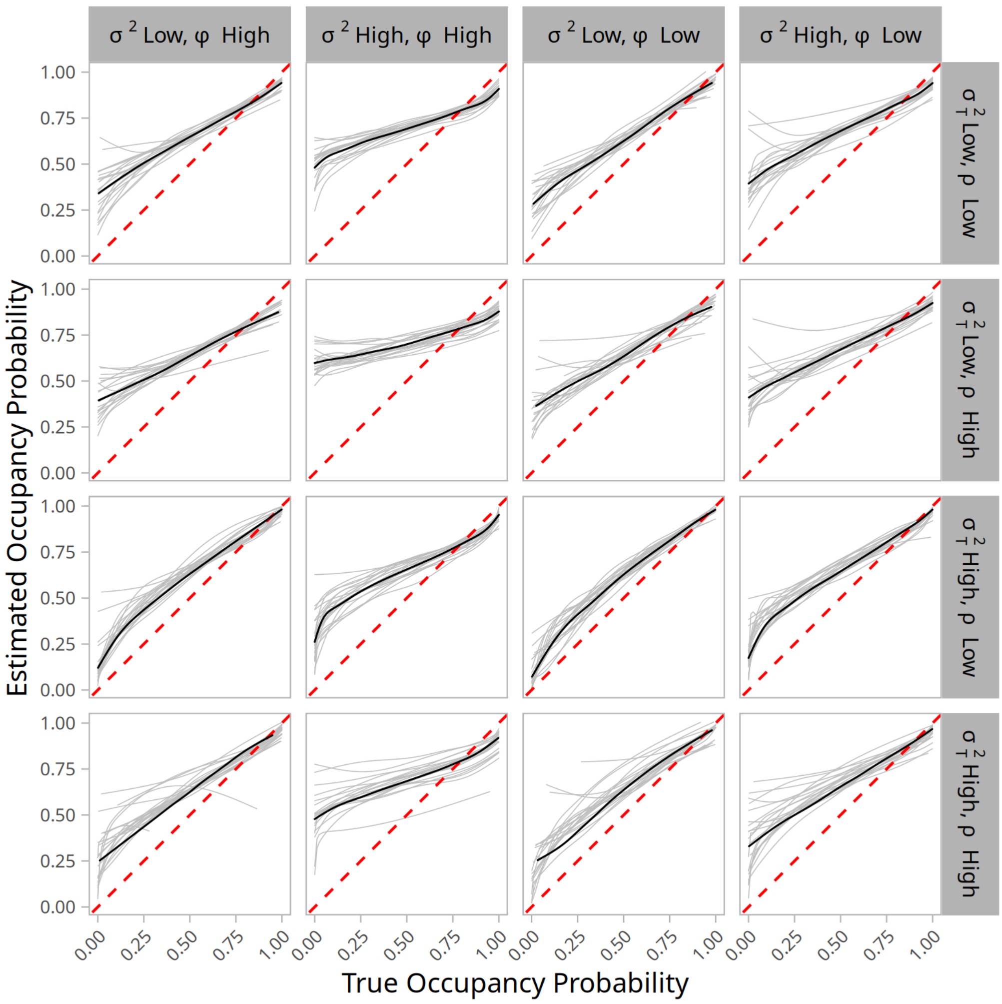

Fig. E.4: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis) under a missispecified occupancy model. Results for the study 3-scenario 3 (observation spot with reestablished amount of data). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 20 simulated datasets, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

The identifiability of $\psi_{it} \times p_{it}$

In this section we also show results of evaluations of the identifiability of the product of occupancy and detection. This analysis is intended to show whether the product is accurately estimated by the model under the different simulated situations. For each simulation run and scenario we obtained the true $\psi_{it} \times p_{it}$ and the estimated $\hat{\psi_{it}} \times \hat{p_{it}}$ product. Note that the detection probability was aggregated over sites and times by taking its average across the J survey occasions.

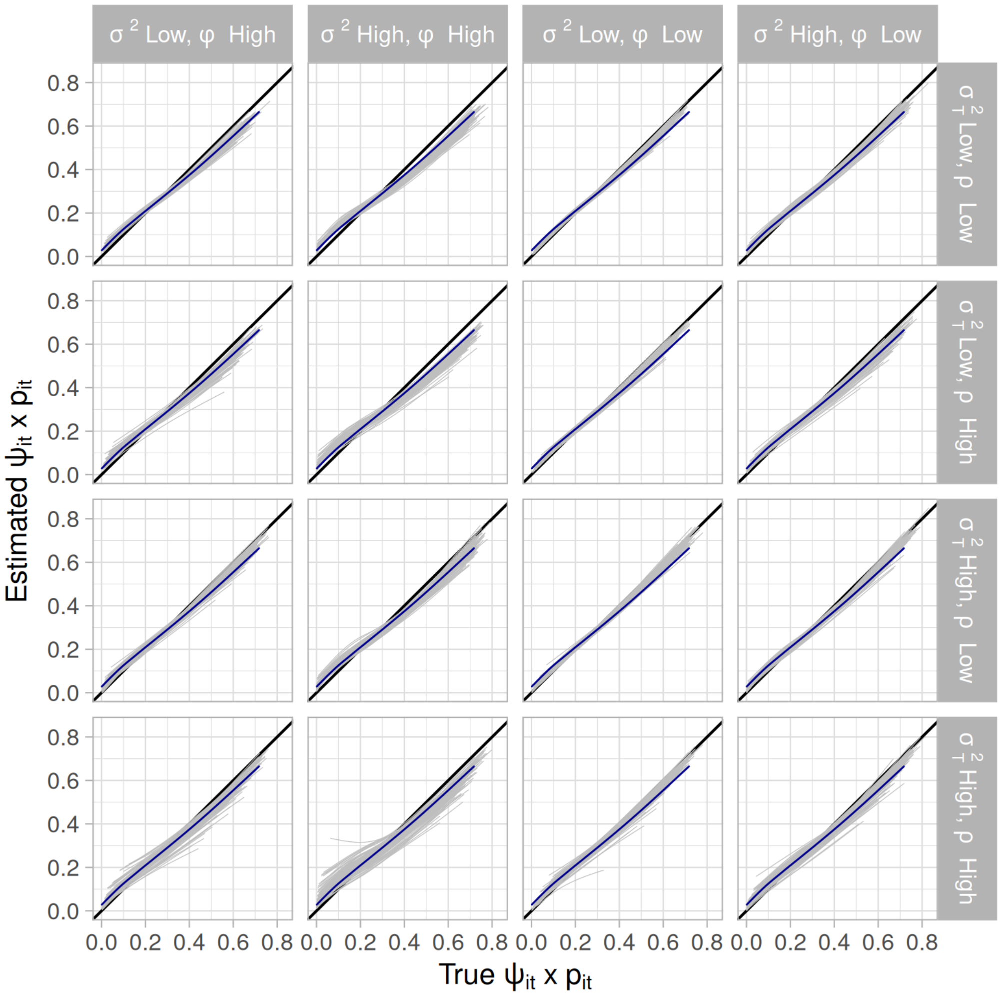

Fig. E.5: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi}_{it}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the baseline scenario study 1-scenario 0. Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi}_{it} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

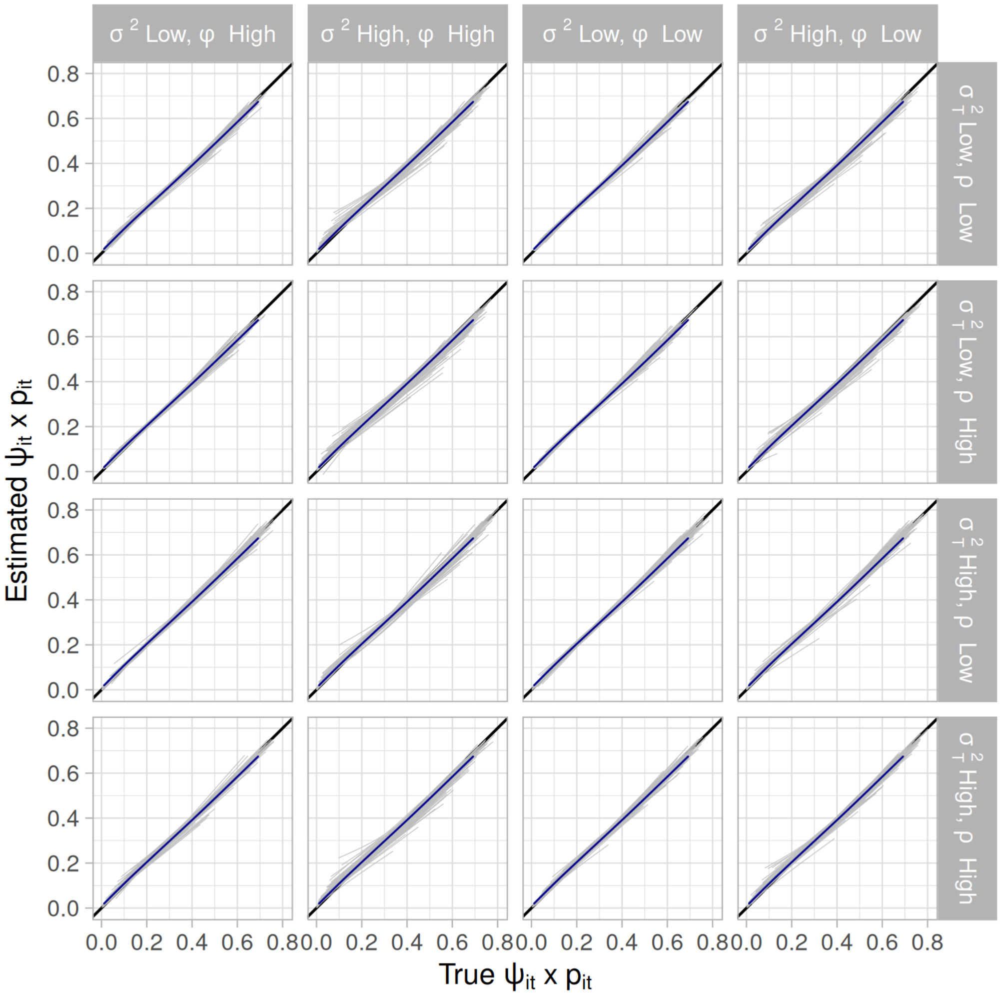

Fig. E.6: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi_{it}}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 1-scenario 1 (smoother spatial autocorrelation). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi_{it}} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

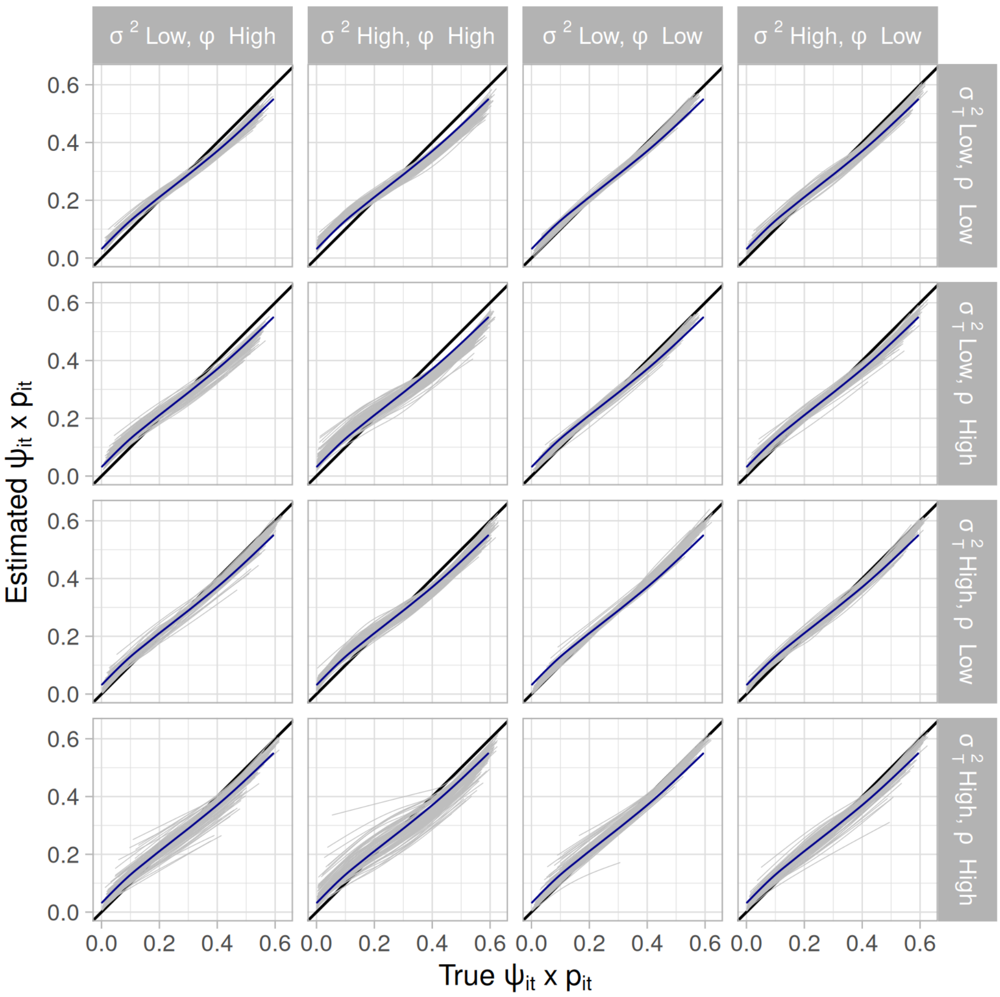

Fig. E.7: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi_{it}}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 2-scenario 0 (Poisson sampling design). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi_{it}} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

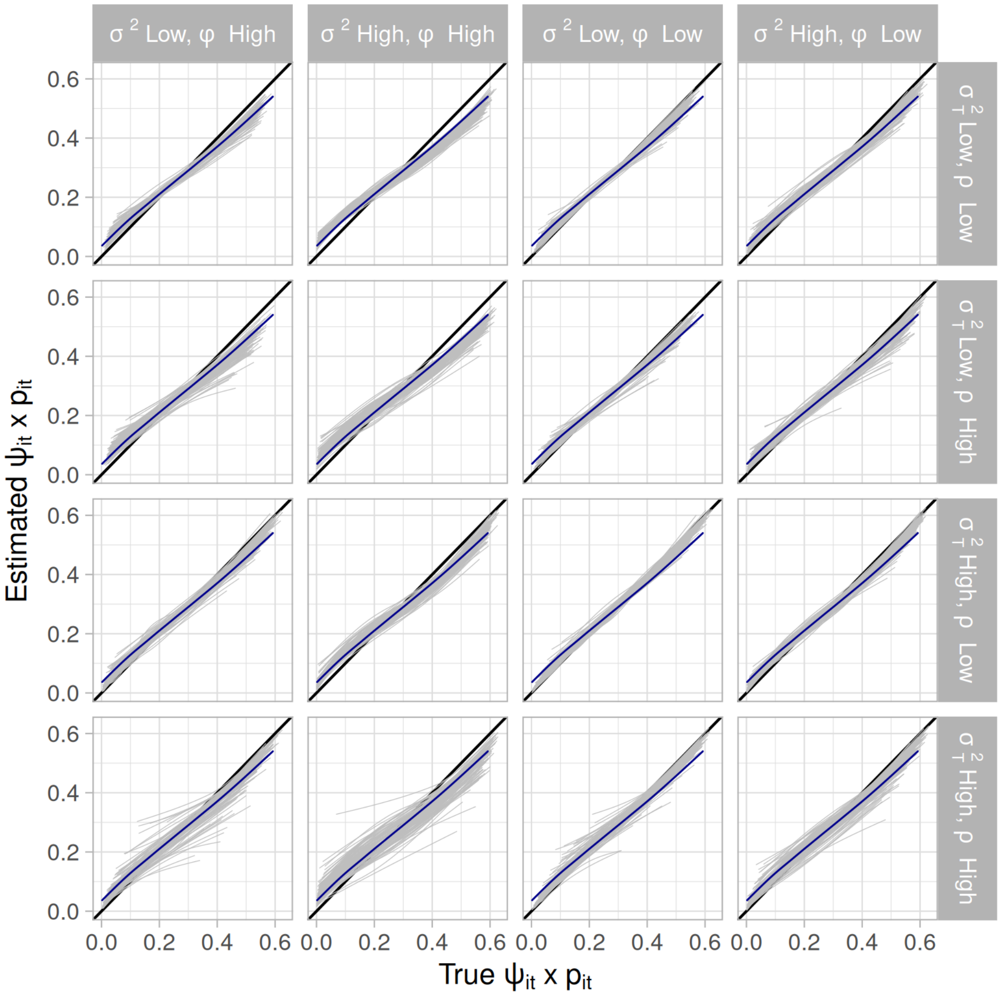

Fig. E.8: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi}_{it}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 2-scenario 1 (Poisson sampling design and latitude as occupancy covariate). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi}_{it} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

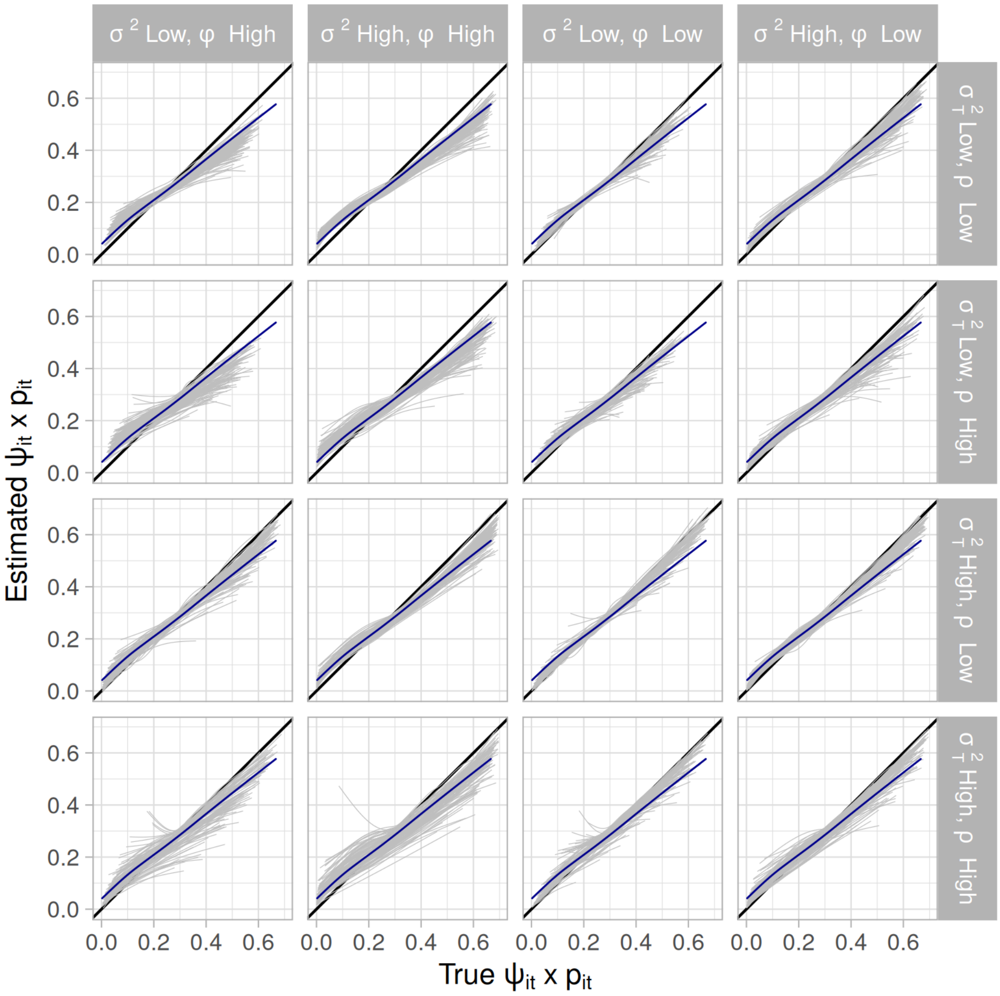

Fig. E.9: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi}_{it}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 2-scenario 2 (Poisson sampling design and overlap of covariates – latitude as occupancy and detection covariate). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi}_{it} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

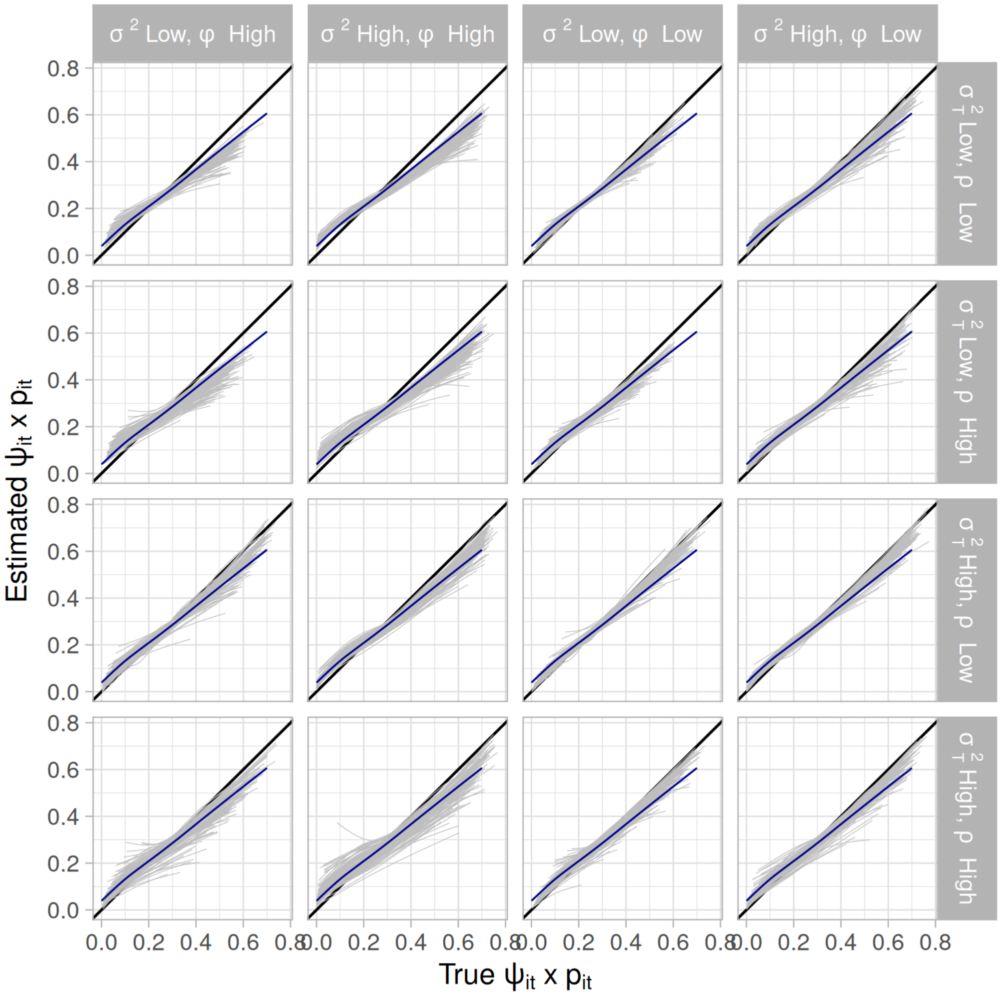

Fig. E.10: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi}_{it}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 2-scenario 3 (Poisson sampling design and partial overlap of covariates). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi}_{it} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

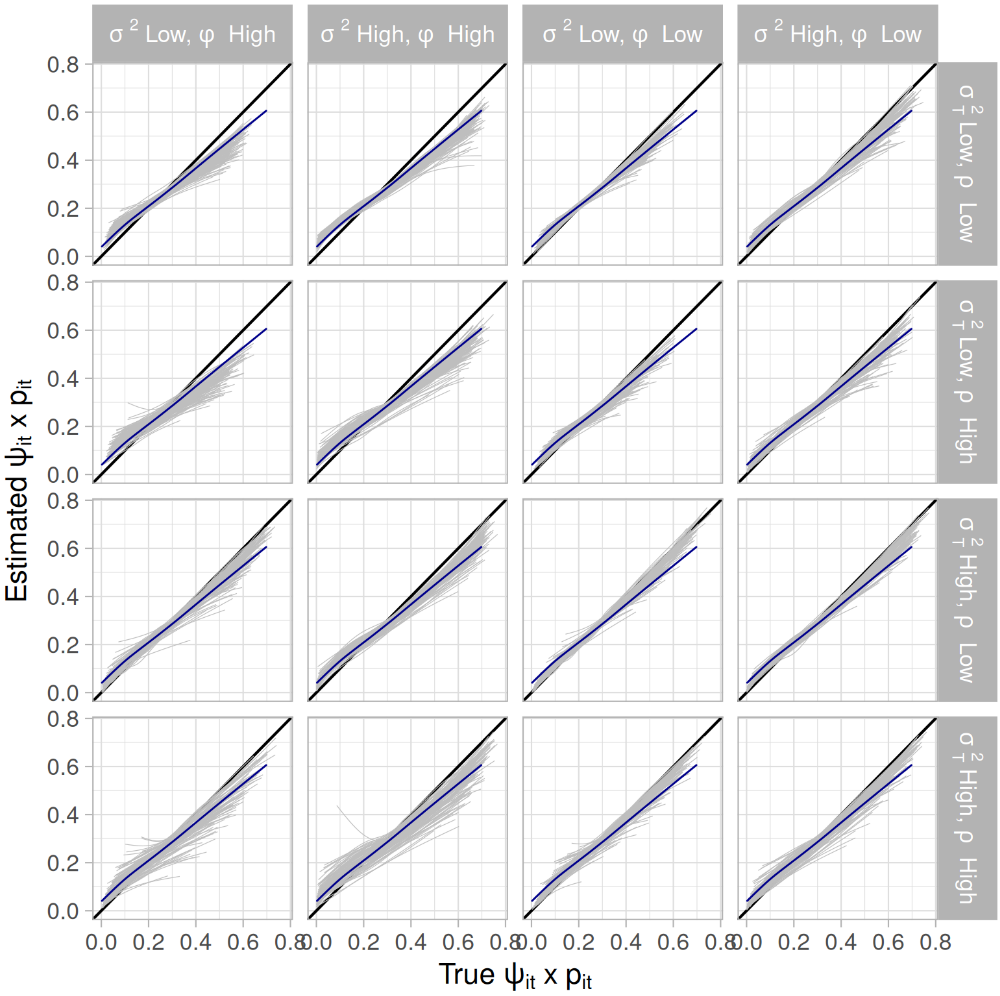

Fig. E.11: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi_{it}}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 3-scenario 1 (Poisson sampling design, partial overlap of covariates, phenology and observer effect). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi_{it}} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

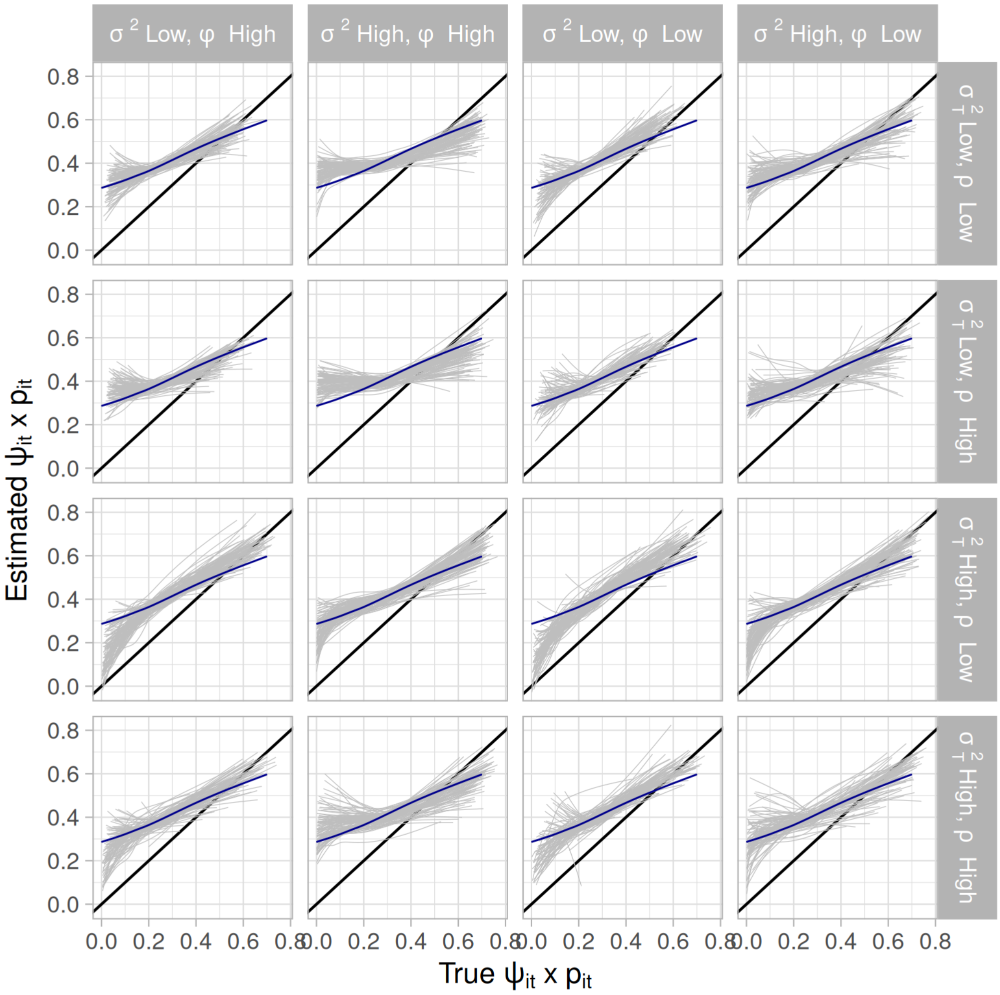

Fig. E.12: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi_{it}}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 3-scenario 2 (Poisson sampling design, partial overlap of covariates, phenology and observer effect, observation spot with 25% of the data). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi_{it}} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

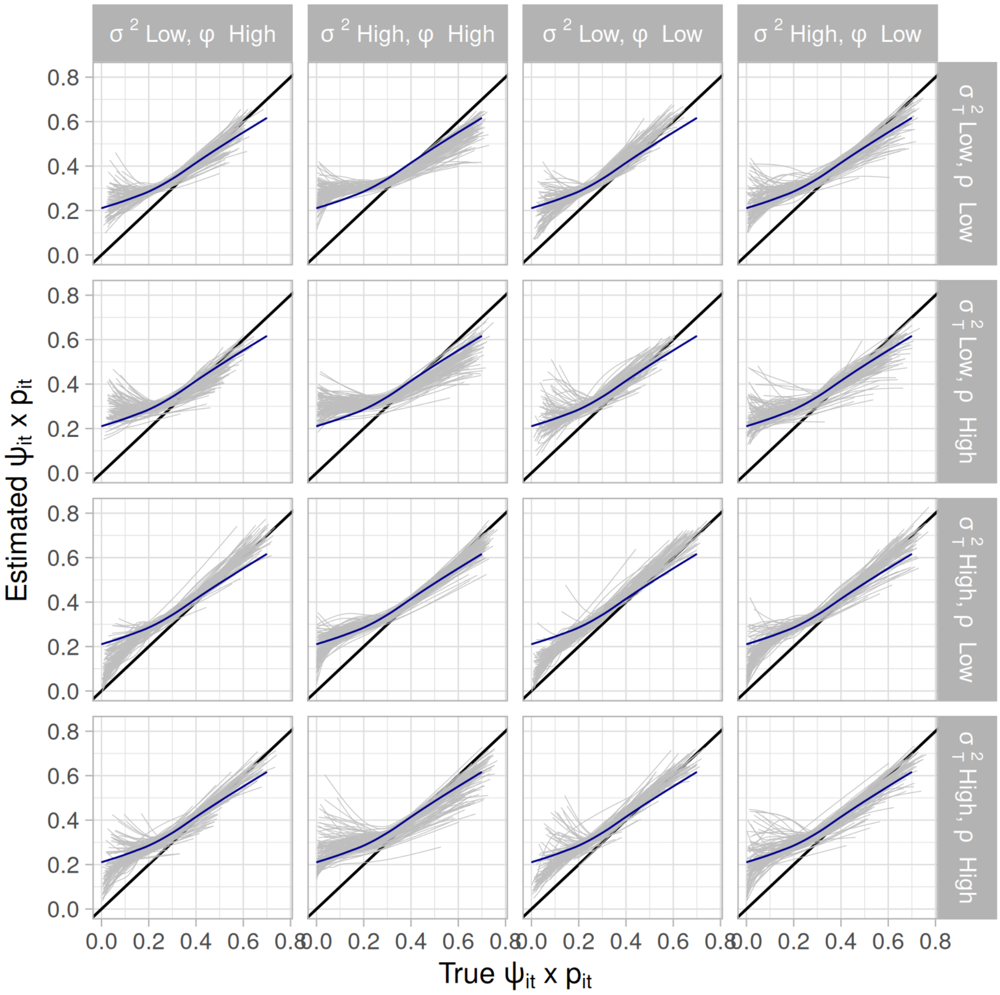

Fig. E.13: Relationship between the product of true site occupancy probability ψ_{it} and detection probability p_{it} (x-axis) and the product of estimated site occupancy probability $\hat{\psi_{it}}$ and detection probability $\hat{p_{it}}$ (y-axis). Results for the study 3-scenario 3 (Poisson sampling design, partial overlap of covariates, phenology and observer effect, observation spot, amount of data reestablished). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between $\psi_{it} \times p_{it}$ and $\hat{\psi_{it}} \times \hat{p_{it}}$ per simulated dataset. The blue line depicts the averaged relationship across the 100 simulated datasets, and the black solid line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

Analysis of simulated data using NNGP = 15 spatial neighbors and observation spot with $I^* = 600$.

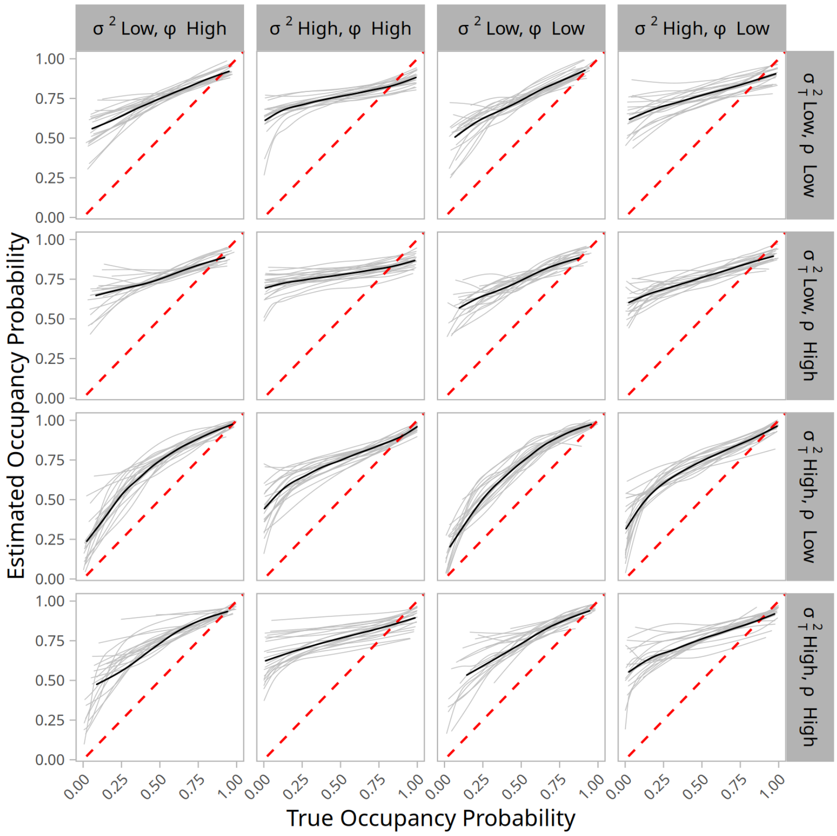

Fig. E.14: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis). The model fitted to occupancy data had NNGP=15 nearest neighbors to approximate the Gaussian Processes. Results for the study 3-scenario 2 (observation spot). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 20 simulated datasets we analyzed, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

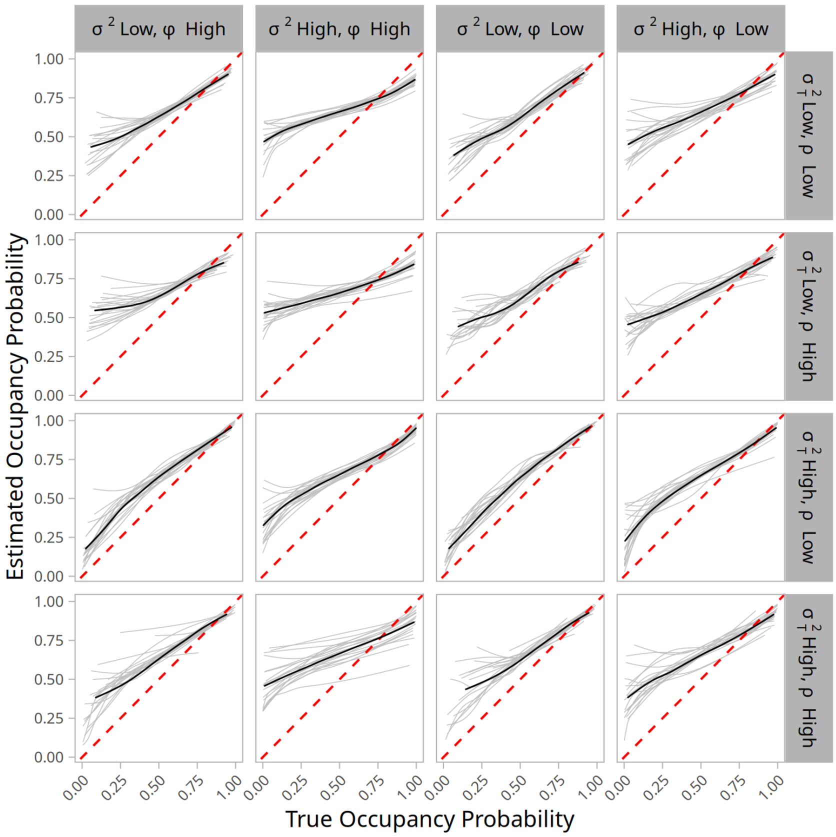

Fig. E.15: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis). The model fitted to occupancy data had NNGP=15 nearest neighbors to approximate the Gaussian Processes. Results for the study 3-scenario 3 (observation spot). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 20 simulated datasets we analyzed, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.

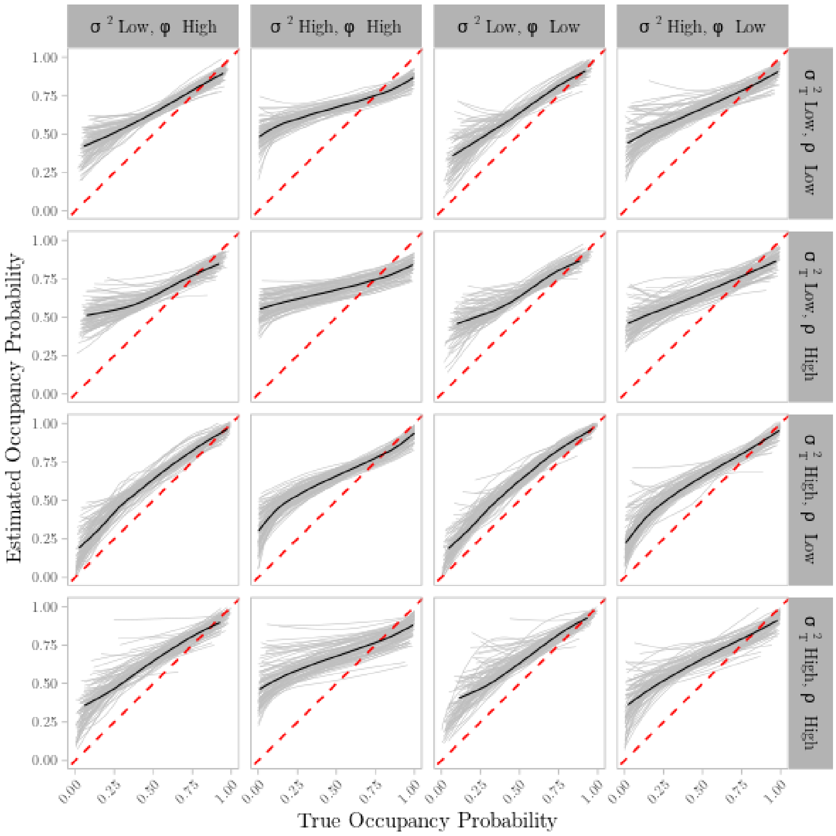

Fig. E.16: Relationship between the true site occupancy probability ψ_{it} (x-axis) and the estimated site occupancy probability $\hat{\psi}_{it}$ (y-axis). The model fitted to occupancy data had NNGP=15 nearest neighbors to approximate the Gaussian Processes. Results for the study 3-scenario 2 with a larger observation spot ($I^* =$ 600 sites). Each gray line represents the locally estimated scatterplot smoothing (LOESS) relationship between ψ_{it} and $\hat{\psi}_{it}$ per simulated dataset. The black line depicts the averaged relationship across the 100 simulated datasets we analyzed, and the red dashed line depicts an 1:1 relationship. Spatial and temporal autocorrelation sub scenarios are represented along columns and rows, respectively.